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# DIFFERENTIATION

## NIMCET PYQ

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By Amit Katiyar (MCA-JNU)

1. If $f(x) = x^n$, then $f(1) + \frac{f'(1)}{1!} + \frac{f''(2)}{2!} \dots \dots \frac{f^n(1)}{n!}$ is equal to

**NIMCET 2007**

(a)0 (b)$2^n$ (c)$2^{n-1}$ (d)$\frac{n(n+1)}{2}$

2. If $f'(x) = g(x)$ and $g'(x) = -f(x)$ for all x and $f(3) = 5 = f'(3)$ then $f''(19) + g''(19)$ is :

**NIMCET 2007**

(a)16 (b)32 (c)64 (d)None

3. If $u = \sec^{-1}\left(\frac{x+1}{x-1}\right) + \sin^{-1}\left(\frac{x-1}{x+1}\right), x \in (0, \infty)$ and $x \neq 1$, then $\frac{dy}{dx}$ is equal to

**NIMCET 2008**

(a)1 (b)$\frac{x-1}{x+1}$ (c)0 (d)$\frac{x+1}{x-1}$

4. If $y = f(x)$ is an odd and differentiable function defined on $(-\infty, \infty)$ such that $f'(3) = -2$, then $f'(-3)$ is equal to

**NIMCET 2009**

(a)4 (b)2 (c)$-2$ (d)0

5. Find $\frac{d}{dx}\left(\sqrt{x} - \frac{5}{\sqrt{x}}\right)$ **NIMCET 2011**

(a)$\frac{1}{2\sqrt{x}} + \frac{3}{2}x^{-3/2}$ (b)$2x - \frac{5}{2}x^{3/2}$
(c)$2x + \frac{5}{2}x^{-3/2}$ (d)None of these

6. If $f(x) = \tan^{-1}\left[\frac{\sin x}{1+\cos x}\right]$, then what is the first derivative of $f(x)$?

**NIMCET 2013**

(a)$\frac{1}{2}$ (b)$-\frac{1}{2}$
(c)2 (d)$-2$

7. If $x = a\cos t, y = b\sin t$, then $\frac{d^2y}{dx^2}$ is

**NIMCET 2013**

(a)$-\frac{b^4}{a^2y^3}$ (b)$\frac{b^4}{a^2x^3}$
(c)$\frac{b}{ay^4}$ (d)$-\frac{a^4}{bx^3}$

8. With the usual notation, $\frac{d^2x}{dy^2}$ is **NIMCET 2015**

(a)$\left(\frac{d^2y}{dx^2}\right)^{-1}$ (b)$\frac{d^2y}{dy^2}\left(\frac{dy}{dx}\right)^{-2}$
(c)$-\left(\frac{d^2y}{dx^2}\right)^{-1}\left(\frac{dy}{dx}\right)^{-2}$ (d)$-\left(\frac{d^2y}{dx^2}\right)\left(\frac{dy}{dx}\right)^{-3}$

9. If $y = \cos^2 x^2$, find $\frac{dy}{dx}$

**NIMCET 2017**

(a)$4x^2 \sin x^2 \cos x^2$
(b)$-4x^2 \cos x^2 \sin x^2$
(c)$2\, x \sin x^2 \cos x^2$
(d)$-2x \cos x^2 \sin x^2$

10. The derivative of $(x^3 + e^x + 3^x + \cot x)$ with represent to x is

**NIMCET 2017**

(a)$3x^2 + e^x + 3^x(\log 3) - \operatorname{cosec}^2 x$
(b)$3x^2 + e^x + 3^x(\log 3) + \operatorname{cosec}^2 x$
(c)$3x^2 + e^x + 3^x(\log 3) - \sec^2 x$
(d)$3x^2 + e^x + 3^x(\log 3) + \sec^2 x$

11. Differentiate $[-\log(\log x), x > 1]$ with respect to x.

**NIMCET 2017**

(a)$\frac{-1}{(x\log x)}$ (b)$\frac{1}{\log x}$
(c)$1/x$ (d)$x\log x$

12. $\frac{d^2y}{dx^2}$ equals

**NIMCET 2018**

(a)$\left(\frac{d^2y}{dx^2}\right)^{-1}$ (b)$-\left(\frac{d^2y}{dx^2}\right)^{-1}\left(\frac{dy}{dx}\right)^{-3}$
(c)$\left(\frac{d^2y}{dx^2}\right)\left(\frac{dy}{dx}\right)^{-2}$ (d)$-\left(\frac{d^2y}{dx^2}\right)\left(\frac{dy}{dx}\right)^{-3}$

13. Differential coefficient of $\log_{10} x$ w.r.t $\log_x 10$ is

**NIMCET 2018**

(a)$-\frac{(\log x)^2}{(\log 10)^2}$ (b)$\frac{(\log_{10} x)^2}{(\log 10)^2}$
(c)$\frac{(\log_x 10)^2}{(\log 10)^2}$ (d)$-\frac{(\log 10)^2}{(\log x)^2}$

14. If $y = \tan^{-1}\left(\frac{3x-x^3}{1-3x^2}\right), -\frac{1}{\sqrt{3}} < x < \frac{1}{\sqrt{3}}$, then $\frac{dy}{dx}$ is

**NIMCET 2021**

(a)$-\frac{1}{1+x^2}$ (b)$\frac{3}{1+x^2}$
(c)$\frac{3}{\sqrt{1+x^2}}$ (d)$\frac{1}{\sqrt{1+x^2}}$

15. If $y = \sin^{-1}\left(\frac{x^2+1}{\sqrt{1+3x^2+x^4}}\right)(x > 0)$, then $\frac{dy}{dx}$ is equals

**NIMCET 2021**

(a)$\frac{x^2-1}{x^4+3x^2+1}$ (b)$\frac{x^2+1}{x^4+3x^2+1}$
(c)$\frac{x^2-1}{x^4-3x^2+1}$ (d)$\frac{x^2+1}{x^4-3x^2+1}$

16. If $x^m y^n = (x+y)^{m+n}$, then $\frac{dy}{dx}$ is

**NIMCET 2022**

(a)$\frac{x+y}{xy}$ (b)xy
(c)x/y (d)y/x

## ANSWER KEY

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 1. | (b) | 2. | (d) | 3. | (c) | 4. | (c) | 5. | (d) |
| 6. | (a) | 7. | (a) | 8. | (d) | 9. | (*) | 10. | (a) |
| 11. | (a) | 12. | (d) | 13. | (a) | 14. | (b) | 15. | (a) |
| 16. | (d) | | | | | | | | |

## Questions Asked in NIMCET 2007 to 2025

| Years | No. of Questions | Years | No. of Questions |
|---|---|---|---|
| NIMCET 2007 | 02 | NIMCET 2016 | 00 |
| NIMCET 2008 | 01 | NIMCET 2017 | 03 |
| NIMCET 2009 | 01 | NIMCET 2018 | 02 |
| NIMCET 2010 | 00 | NIMCET 2019 | 00 |
| NIMCET 2011 | 01 | NIMCET 2020 | 00 |
| NIMCET 2012 | 00 | NIMCET 2021 | 02 |
| NIMCET 2013 | 02 | NIMCET 2022 | 01 |
| NIMCET 2014 | 00 | NIMCET 2023 | 00 |
| NIMCET 2015 | 01 | NIMCET 2024 | 00 |
| | | NIMCET 2025 | 00 |

## SOLUTION

1. **(b)** f(x) = $x^n$ then f(1) + $\frac{f'(1)}{1!}+\frac{f''(1)}{2!}\ldots\ldots\ldots\frac{f^n(1)}{n!}$

f(x) = $x^n$ f(1) = 1

f′(x) = $nx^{n-1}$ f′(1) = n

f''(x) = n (n–1) $x^{n-2}$ f''(1) = n (n–1)

$f^n(x)$ = n (n–1)(n–2) ......1 $f^n(1)$ = n!

$= f(1)+\frac{f'(1)}{1!}+\frac{f''(1)}{2!}+\ldots\ldots\ldots\ldots\frac{f^n(1)}{n!}$

$= 1+\frac{n}{1!}+\frac{n(n-1)}{2!}+\frac{n(n-1)(n-2)}{3!}+\cdots\ldots\frac{n!}{n!}$

= $^nC_0$ + $^nC_1$ + $^nC_2$ + ......$^nC_n$

= $2^n$

2. **(d)**

$f'(x)=g(x)$ $g'(x)=-f(x)$

$f(3)=5=f'(3)$, then $f''(19)+g''(19)$

Let h(x) = f''(x) + g''(x)

$h'(x)=2f(x)\cdot f'(x)+2g(x)\cdot g'(x)$

f' (x) = 0

It means h(x) is a constant function.

f''(x) + g'' (x) = constant

f'' (3) + g'' (3) = $5^2 + 5^2$ = 50

f'' (19) + g'' (19) = 50

3. **(c)** $y=\sec^{-1}\left(\frac{x+1}{x-1}\right)+\sin^{-1}\left(\frac{x-1}{x+1}\right), x\in(0,\infty)$

$y=\cos^{-1}\left(\frac{x-1}{x+1}\right)+\sin^{-1}\left(\frac{x-1}{x+1}\right)$

$\therefore\ y=\frac{\pi}{2}$

$\left[\because \sin^{-1}x+\cos^{-1}x=\frac{\pi}{2}\right]$ $\Rightarrow\frac{dy}{dx}=0$

4. **(c)** $y=f(x)$

$f(-x)=-f(x)$ [As $f(x)$ is odd function]

$\Rightarrow -f'(-x)=-[f'(x)]$

$\Rightarrow\ f'(-x)=[f'(x)]$

Hence, $f'(x)$ will be an even function.

$f'(3)=f'(-3)=-2$

5. **(d)** $\frac{d}{dx}\left[\sqrt{x}-\frac{5}{\sqrt{x}}\right]=\frac{1}{2\sqrt{x}}-5\frac{d}{dx}(x)^{-1/2}$

$=\frac{1}{2}(x)^{-1/2}+\frac{5}{2}x^{-3/2}$

6. **(a)** $f(x)=\tan^{-1}\left[\frac{\sin x}{1+\cos x}\right]=\tan^{-1}\left[\frac{2\sin\frac{x}{2}\cos\frac{x}{2}}{2\cos^2\frac{x}{2}}\right]$

$f(x)=\tan^{-1}\left[\tan\left(\frac{x}{2}\right)\right]=\frac{x}{2}\Rightarrow f'(x)=\frac{1}{2}$

7. **(a)** $x=a\cos t, y=b\sin t$

$\frac{dx}{dt}==-a\sin t,\frac{dy}{dt}=b\cos t\Rightarrow\frac{dy}{dx}=\frac{dy/dt}{dx/dt}=-\frac{b}{a}\cot(t)$

$\frac{d^2y}{dx^2}=\frac{d}{dx}\left(\frac{dy}{dx}\right)=\frac{d}{dt}\left(\frac{dy}{dx}\right)\frac{dt}{dx}=\left(-\frac{b}{a}\right)(-\mathrm{cosec}^2\,t)\left(\frac{dt}{dx}\right)$

$=\frac{b}{a}(\mathrm{cosec}^2\,t)\left(-\frac{1}{a\sin t}\right)$

$=-\left(\frac{b}{a^2}\right)\mathrm{cosec}^3\,t=-\frac{b}{a^2\sin^3 t}$

$=-\frac{b^4}{a^2y^3}$ (as $\sin t=\frac{y}{b}$)

8. **(d)** Since, $\frac{dx}{dy}=\frac{1}{\frac{dy}{dx}}=\left(\frac{dy}{dx}\right)^{-1}$

$\Rightarrow\frac{d^2x}{dy^2}=\frac{d}{dy}\left(\frac{dx}{dy}\right)=\frac{d}{dx}\left(\frac{dy}{dx}\right)^{-1}\frac{dx}{dy}$

$\Rightarrow\frac{d^2x}{dy^2}=-\left(\frac{d^2y}{dx^2}\right)\left(\frac{dy}{dx}\right)^{-2}\left(\frac{dx}{dy}\right)$

$\Rightarrow\left(\frac{d^2x}{dy^2}\right)=-\left(\frac{d^2y}{dx^2}\right)\left(\frac{dy}{dx}\right)^{-3}$

9. **(Wrong)** $=y=\cos^2x^2$

$\frac{dy}{dx}=2\cos(x^2)(-\sin x^2)(2x)=-4x\cos x^2\sin x^2$

10. **(a)**

$f(x)=x^3+e^x+3^x+\cot x$

$f'(x)=3x^2+e^x+3^x\log 3-\mathrm{cosec}^2\,x$

11. **(a)**

$y=[-\log_e(\log_e x), x>1]$

On differentiation,

$\frac{dy}{dx}=-\frac{d}{dx}(\log(\log x))=-\frac{1}{\log x}\frac{d}{dx}(\log x)=-\frac{1}{x\log x}$

12. **(d)** Refer to solution 8.

13. **(a)** Let, $u=\log_{10}x$ and $v=\log_x 10$

$\therefore\ u=\frac{\log_e x}{\log_e 10}, v=\frac{\log_e 10}{\log_e x}$

$\frac{du}{dx}=\frac{1}{\log_e 10}\cdot\frac{1}{x}$ [differentiating w.r.t. $x$ ]

and $\frac{dv}{dx}=-\log_e 10\cdot\frac{1}{(\log_e x)^2}\cdot\frac{1}{x}$

Now, $\frac{du}{dv}=\frac{du/dx}{dv/dx}=\frac{\frac{1}{\log_e 10}\cdot\frac{1}{x}}{-\frac{\log_e 10}{(\log_e x)^2}\cdot\frac{1}{x}}$

$=-\frac{1}{\log_e 10}\cdot\frac{1}{x}\cdot\frac{x}{\log_e 10}(\log_e x)^2=-\frac{(\log x)^2}{(\log 10)^2}$

**14.** **(b)** $y = \tan^{-1}\left[\frac{3x-x^3}{1-3x^2}\right], \left[-\frac{1}{\sqrt{3}} < x < \frac{1}{\sqrt{3}}\right]$

$y = \tan^{-1}(\tan 3\theta) = 3\theta$

Put $x = \tan\theta \Rightarrow \theta = \tan x^{-1}\left[\because \tan 3\theta = \frac{3\tan\theta - \tan^3\theta}{1-3\tan^2\theta}\right]$

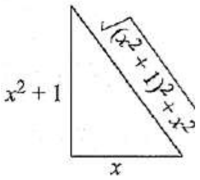
$y = 3\tan^{-1} x$

$\Rightarrow \frac{dy}{dx} = = \frac{3}{1+x^2}$

**15.** **(a)** $y = \sin^{-1}\left(\frac{x^2+1}{\sqrt{1+3x^2+x^4}}\right), x > 0$

$y = \sin^{-1}\left[\frac{x^2+1}{\sqrt{(x^2+1)^2+x^2}}\right]$

$y = \tan^{-1}\left(\frac{x^2+1}{x}\right)$




$y = \tan^{-1}\left(\frac{x^2+1}{x}\right)$

$\frac{dy}{dx} = \frac{1}{1+\left(\frac{x^2+1}{x}\right)^2} \times \frac{x(2x)-(x^2+1)}{x^2} = \frac{x^2-1}{x^4+3x^2+1}$

**16.** **(d)** $x^m y^n = (x+y)^{m+n}$

Taking log both sides,

$m\ln x + n\ln y = (m+n)\ln(x+y)$

On differentiating

$\frac{m}{x} + \frac{n}{y}\frac{dy}{dx} = \frac{m+n}{(x+y)}\left(1 + \frac{dy}{dx}\right)$

$\Rightarrow \left(\frac{n}{y} - \frac{m+n}{x+y}\right)\frac{dy}{dx} = \frac{m+n}{x+y} - \frac{m}{x}$

$\Rightarrow \left(\frac{n(x+y)-y(m+n)}{y(x+y)}\right)\frac{dy}{dx} = \frac{(m+n)x-m(x+y)}{x(x+y)}$

$\Rightarrow \left(\frac{nx-my}{y}\right)\frac{dy}{dx} = \left(\frac{nx-my}{x}\right)$

$\Rightarrow \frac{dy}{dx} = \frac{y}{x}$